

# Peridot: Artifact Guide

## Middleware 2026 Artifact Evaluation

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### Artifact for ACM Middleware '26.

Peridot is an I/O-extensible WebAssembly runtime that lets the implementation of individual WASI hostcalls be replaced at run time by plugins, without recompiling or modifying the guest module. This artifact packages the runtime, the Overseer control plane, every context, the guest applications and an object store into a single container image, with one script per paper figure and one plotting script per figure.

# 1 What this artifact contains

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|             |                                                                                                                                            |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Source      | The runtime, the Overseer, every context, the containerd shim, and the ImageNet workload.                                                  |
| Build       | <code>artifact/Dockerfile</code> and <code>artifact/docker-compose.yml</code> , which also starts an S3-compatible object store.           |
| Guests      | Precompiled <code>wasm32-wasip1</code> modules in <code>artifact/guests/</code> ; <code>artifact/bin/build-guests.sh</code> rebuilds them. |
| Experiments | One script per figure in <code>artifact/experiments/</code> , each writing a CSV to <code>artifact/results/</code> .                       |
| Analysis    | One matplotlib script per figure in <code>artifact/plots/</code> , writing PDF and PNG to <code>artifact/figures/</code> .                 |

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## 2 Badges claimed

**Artifacts Available, Artifacts Functional, and Results Reproduced.**

Results Reproduced covers Figures 3, 5, 6 and 7, and the two rows of Table 1 that need neither a native IBM GEDS installation nor a FUSE mount. Figure 4 is not reproduced. See [What this artifact does and does not reproduce](#).

### 2.1 Availability

Source repository: <https://github.com/pablogs98/Peridot>

Licence: [Apache License 2.0](#). Third-party material that this licence does not cover is listed in [THIRD-PARTY-NOTICES.md](#).

**Before submission, fill in the line below and delete this note.** The Artifacts Available badge asks for a public, archival location. A GitHub URL alone is not archival, because history can be rewritten or the repository removed; mint a Zenodo DOI from the tagged release and cite it here.

- Archived release (DOI): `TODO`

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## 3 System requirements

Docker and Docker Compose. Four cores, 8 GiB of RAM and about 10 GB of free disk. On Docker Desktop the disk counts against the VM's limit, not the host's. For timings comparable to the paper, use an x86-64 Linux host.

The paper's numbers were measured on AWS EC2 running Ubuntu 22.04: an `m7i.2xlarge` (8 vCPU, 32 GiB) for §4.1, and a `t3.xlarge` (4 vCPU, 16 GiB, 5 Gbps) with Amazon S3 for §4.2 and §4.3. This artifact uses a local MinIO container instead, at reduced scale.

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## 4 Getting Started Guide

About 10 minutes, most of it download. Confirms the artifact builds, runs and produces figures. No experiment is run to completion; that is [Step-by-Step Instructions](#).

### 4.1 1. Build the image and start the object store

```
cd <repository root>
docker compose -f artifact/docker-compose.yml up -d
```

Builds the runtime, the Overseer, every context, the plugin and the external batcher from source, and starts MinIO with the bucket created.

## 4.2 2. Confirm the runtime and a context chain work

```
docker compose -f artifact/docker-compose.yml run --rm peridot experiments/smoke.sh
```

Expected output ends with:

```
[INFO peridot] Context chain: ["counter"]
[INFO peridot_counter_ctx] Total bytes written: 67108982
==> PASS - counter intercepted the write path, 67108982 bytes counted
```

The last digits vary: the total is the guest's 64 MiB payload plus its own progress lines.

## 4.3 3. Confirm the analysis path works

```
docker compose -f artifact/docker-compose.yml run --rm peridot \
python3 plots/plot_all.py --reference
```

Regenerates all four figures from the sample CSVs in `results/reference/`, writing PDF and PNG into `artifact/figures/` on the host. They correspond to Figures 3, 5, 6 and 7 of the paper, and each script prints the relevant ratios to stdout.

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# 5 Step-by-Step Instructions

Run these from `/artifact` inside the container:

```
docker compose -f artifact/docker-compose.yml exec peridot bash
```

| Claim                                                       | Script (in <code>experiments/</code> ) | Runtime |
|-------------------------------------------------------------|----------------------------------------|---------|
| §3.2, Fig. 3: interposition costs essentially nothing       | <code>exp_overhead.sh</code>           | ~2 min  |
| §4.1, Fig. 5: bandwidth tracks declared demand              | <code>exp_io.sh</code>                 | ~6 min  |
| §4.2, Tab. 1: async offload beats sync offload              | <code>exp_storage.sh</code>            | ~2 min  |
| §4.2, Fig. 6: in-runtime batching beats an external batcher | <code>exp_storage.sh</code>            | ~8 min  |
| §4.3, Fig. 7: hostcall batching raises write IOPS           | <code>exp_hostcall_batching.sh</code>  | ~2 min  |

## 5.1 Running everything at once

```
docker compose -f artifact/docker-compose.yml run --rm peridot \
experiments/run_all.sh
```

About 15 minutes at reduced scale. Runs every experiment, regenerates every figure, and prints a summary of what ran and what was skipped. CSVs land in `artifact/results/` and figures in `artifact/figures/` on the host.

## 5.2 Scale

Experiments run a reduced configuration by default. `PERIDOT_AE_FULL=1` switches to the paper's parameters, which need the paper's hardware and run for hours:

```
docker compose -f artifact/docker-compose.yml run --rm \
-e PERIDOT_AE_FULL=1 peridot experiments/run_all.sh
```

The reduced configuration lowers volumes, iteration counts and repetitions. Each script states both parameters at the top.

### 5.3 Figure 3: interposition overhead (§3.2)

```
experiments/exp_overhead.sh          # ~2 min
python3 plots/plot_overhead.py
```

Times `fd_write`, `fd_pwrite`, `fd_read` and `fd_pread` at 1 KiB inside the guest, with an empty context chain and with `syscall-batching` at `num_writes: 1`, which disables batching and leaves only interposition.

**Expected.** Bars of equal height within the error bars; the printed percentages straddle zero. Observed -0.8% to +6.9%.

### 5.4 Figure 5: dynamic I/O provisioning (§4.1)

```
experiments/exp_io.sh                # ~6 min
python3 plots/plot_io.py
```

Four colocated modules write concurrently under a global 1 Gbps policy with demands of 100/200/300/400 Mbps, once without the Overseer (each pinned to an equal quarter) and once with it.

**Expected.** Upper subplot: the four lines overlap regardless of demand. Lower subplot: they separate in proportion to it. Observed 224/225/229/227 Mbps against 74/174/280/378 Mbps.

At the paper's demands the total (100+200+300+400) equals the policy exactly, so max-min fair share grants each module its demand and a module that finishes frees capacity no other module is asking for. The allocation is therefore flat and demand-proportional rather than stepping up over time. Oversubscribing the policy shows the reallocation:

```
IO_DEMANDS_MBPS="100 900 900 900" experiments/exp_io.sh
```

Each module is then clamped below its demand, and the clamps loosen as modules finish. The policy is unit-tested in `crates/peridot-overseer/src/policy.rs` (`cargo test -p peridot-overseer`).

### 5.5 Table 1: asynchronous versus synchronous offload (§4.2)

```
experiments/exp_storage.sh          # ~2 min for this part
python3 plots/plot_batch.py
```

Runs the ImageNet guest, which only opens, writes and closes ordinary files, with the `s3` context in asynchronous mode (uploads spawned) and in synchronous mode (`sync: true`, each upload awaited inside `fd_write`).

**Expected.** `s3-async-ctx` ahead of `s3-sync-ctx`. Observed 107 vs 49 MB/s, a factor of 2.2 against the paper's 5.1x on Amazon S3; `artifact/env.example` shows how to point at a remote store. The `s3fs` and `geds-ctx` rows are not run, see [Not reproduced](#).

### 5.6 Figure 6: in-runtime Parquet batching (§4.2)

```
experiments/exp_storage.sh          # ~8 min for this part
python3 plots/plot_batch.py
```

The same guest at batch sizes 2 to 128. The Peridot arm uses the `batch` context, which groups files and uploads each group as one Parquet object from inside the runtime; the Wasmtime arm runs with no context and a separate `tensor_batcher` process. Both arms are timed until every batch is durably in the object store, since the external batcher keeps working after the guest exits.

**Expected.** Peridot ahead at every batch size, largest margin at the smallest. Observed 2.05x at batch 2 and 1.37x at batch 128, against the paper's 1.15x at the smallest batch.

### 5.7 Figure 7: WASI hostcall batching (§4.3)

```
experiments/exp_hostcall_batching.sh # ~2 min
python3 plots/plot_hostcall_batching.py
```

Sequential writes of 64 B, 256 B, 1 KiB and 4 KiB while the `syscall-batching` context buffers `num_writes` of them and flushes each group as one host write, sweeping `num_writes` from 1 (batching off) to 1024.

**Expected.** IOPS rise with batch size, most at 64 B, least at 4 KiB, flattening past 128. The multiplier scales with the cost of a host `write`, so it is far larger inside a VM than the paper’s 4.90x on EC2.

---

## 6 Interpreting the output

Every experiment produces three things.

1. **A CSV in `results/`,** one row per measurement; aggregation happens in the plot script.

|                                    |                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------|
| <code>overhead.csv</code>          | <code>arm,rep,op,bytes,iterations,</code><br><code>mean_us,p50_us,p99_us,stddev_us</code>    |
| <code>io.csv</code>                | <code>arm,module,demand_mbps,t_seconds,written_bytes</code>                                  |
| <code>storage_table1.csv</code>    | <code>setup,rep,images,seconds,throughput_mbs</code>                                         |
| <code>batch.csv</code>             | <code>arm,batch_size,rep,images,seconds,throughput_mbs</code>                                |
| <code>hostcall_batching.csv</code> | <code>io_size,batch_size,writes_per_run,runs,</code><br><code>mean_kiops,stddev_kiops</code> |

`io.csv` records cumulative bytes; the plot script differentiates them into a bandwidth series. The `arm` or `setup` column separates the baseline from Peridot and is the comparison each figure makes.

2. **A figure in `figures/`,** as both PDF and PNG.

3. **The numbers the claim rests on, printed to `stdout`.** For example `plots/plot_hostcall_batching.py` prints:

```
speedup over no batching (num_writes = 1):
    64 B: baseline      16.0 K -> best      8236.5 K at num_writes=1024  (513.53x)
```

and `plots/plot_batch.py` prints Table 1 in the paper’s layout. Compare the ordering and trend against the paper rather than the magnitudes. `plots/plot_all.py` reports which figures it skipped and why.

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## 7 What this artifact does and does not reproduce

### 7.1 Reproduced

- **Figure 3**, interposition overhead.
- **Figure 5**, dynamic I/O provisioning, including the Overseer’s max-min fair share loop: modules register a demand, the Overseer computes `token_rate`, and each module’s token bucket applies it. Modules deregister on exit and the policy recomputes.
- **Figure 6**, in-runtime Parquet batching.
- **Figure 7**, hostcall batching.
- **Table 1**, the `s3-sync-ctx` and `s3-async-ctx` rows.

Details, including expected output, are under [Step-by-Step Instructions](#).

### 7.2 Not reproduced

- **geds-ctx (Table 1)** needs a native IBM GEDS installation. Source is in `crates/contexts/peridot-geds-ctx/`; [docs/peridot.md](#) documents the build.
  - **s3fs (Table 1)** needs FUSE and a privileged container. It is the row the paper normalises to, so `plots/plot_batch.py` normalises to the slowest arm present.
  - **Figure 4**, Overseer scalability, needs two nodes of a specific size, and the `UpdateMetrics` RPC carries no timestamp, so there is no latency instrumentation to read.
- 

## 8 Layout

`artifact/`

|                    |                                                  |
|--------------------|--------------------------------------------------|
| Dockerfile         | multi-stage build, invoked by docker compose     |
| docker-compose.yml | Peridot + MinIO                                  |
| env.example        | copy to .env to override credentials or endpoint |
| guests/            | prebuilt .wasm modules                           |
| configs/           | example Peridot configuration files              |
| experiments/       | one script per figure, each writing a CSV        |
| plots/             | one matplotlib script per figure                 |
| results/           | CSVs, mounted from the host                      |
| results/reference/ | sample data, for --reference                     |
| figures/           | generated PDFs and PNGs, mounted from the host   |
| bin/               | guest and PDF build scripts                      |

## 8.1 Guest modules

| Module                      | Source                                 | Used by              |
|-----------------------------|----------------------------------------|----------------------|
| hostcall_latency.wasm       | c-wasm/hostcall_latency                | Figure 3             |
| io_writer.wasm              | c-wasm/io_writer                       | Figure 5, smoke test |
| iops.wasm                   | c-wasm/iops                            | Figure 7             |
| imagenet-preprocessing.wasm | crates/contexts/imagenet-preprocessing | Table 1, Figure 6    |
| helloworld.wasm             | c-wasm/helloworld                      | ad-hoc use           |

bin/build-guests.sh rebuilds them from source. It needs wasi-sdk and is not required to run the experiments.

## 9 Running without Docker

**Linux only.** The Overseer’s metrics producer reads /proc/<pid>/io, so on a host without /proc its thread dies on the first tick and each module keeps the rate from its own configuration file. The Figure 5 plot then does not demonstrate the Overseer. Other experiments are unaffected.

Needs the dependencies in docs/peridot.md (protoc, cmake, a C toolchain):

```
cargo build --release --locked -p peridot-cli -p peridot-overseer \
    -p tensor_batcher -p peridot-trace-plugin
pip install -r artifact/plots/requirements.txt boto3

cd artifact
experiments/smoke.sh
S3_ENDPOINT=http://127.0.0.1:9000 AWS_ACCESS_KEY_ID=... \
    AWS_SECRET_ACCESS_KEY=... experiments/run_all.sh
```

Build the runtime and any plugin **in one cargo invocation**, as above. Contexts cross the plugin boundary as ordinary Rust types with no stable ABI, and Cargo resolves features per invocation, so separate builds can disagree and the plugin will fail to load.

## 10 The containerd shim

containerd-shim-peridot runs a Wasm workload as an ordinary container under ctr, nerdctl or Kubernetes. It cannot run inside this image: it needs a privileged Linux host with containerd and a registry. crates/containerd-shim-peridot/README.md has the walkthrough.

## 11 Acknowledgements



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